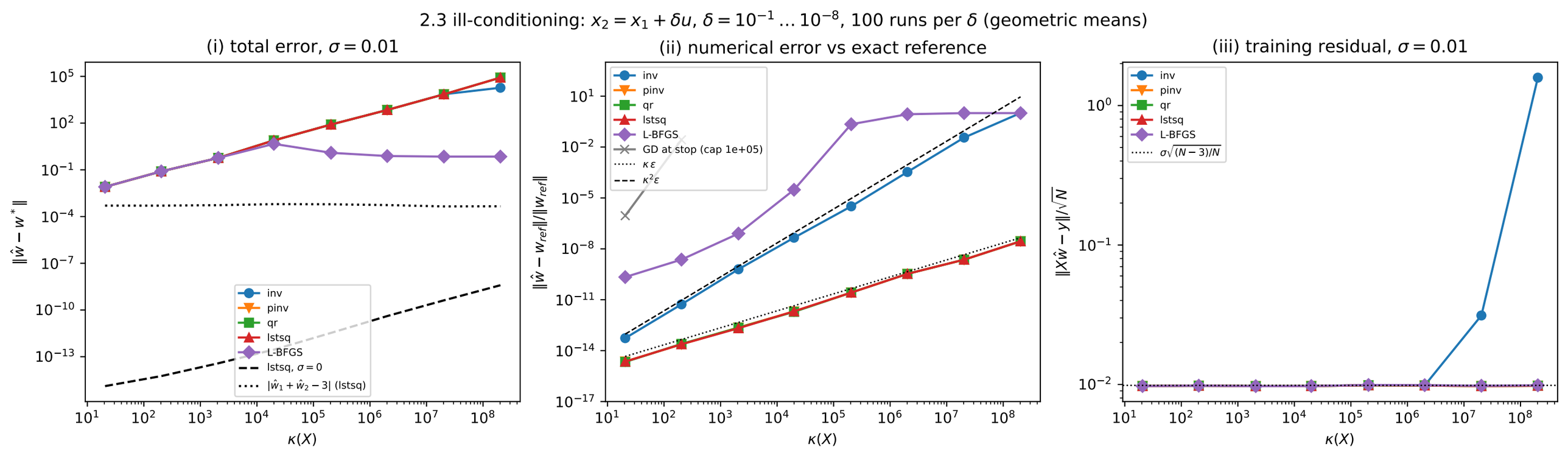
2.3 ill-conditioning: $x_2 = x_1 + \delta u$, $\delta = 10^{-1} \ldots 10^{-8}$, 100 runs per δ (geometric means)
(i) total error, $\sigma = 0.01$
$\|\hat{w} - w^*\|$
$\kappa(X)$
inv
pinv
qr
lstsq
L-BFGS
lstsq, $\sigma = 0$
$|\hat{w}_1 + \hat{w}_2 - 3|$ (lstsq)
(ii) numerical error vs exact reference
$\|\hat{w} - w_{ref}\|/\|w_{ref}\|$
$\kappa(X)$
inv
pinv
qr
lstsq
L-BFGS
GD at stop (cap 1e+05)
$\kappa\,\varepsilon$
$\kappa^2\varepsilon$
(iii) training residual, $\sigma = 0.01$
$\|X\hat{w} - y\|/\sqrt{N}$
$\kappa(X)$
inv
pinv
qr
lstsq
L-BFGS
$\sigma\sqrt{(N-3)/N}$